GIS-Based Site Selection for Green Infrastructure

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1. Introduction

In the past years, GI has received growing attention not only for its potential to improve public health and urban climate adaptation but also to alleviate social inequality.

This assessment takes Greater Manchester as the research area and uses the GIS - based Multi - Criteria Evaluation (MCE) and Weighted Linear Combination (WLC) methods to systematically identify and evaluate potential green infrastructure intervention sites. On this basis, the study further designs two types of scenario plans with different planning objectives oriented towards environmental benefits, focusing on urban thermal environment regulation and ecological benefits. The other is the "site selection for the largest community garden" oriented towards humanistic and social benefits, aiming to address the issues of green space accessibility and social needs at the community level. Through the above intervention plans, the study attempts to alleviate the social - environmental inequality caused by the uneven distribution of urban green space.

2. Methodology

2.1 Analytical Framework and Logic Model

The Multi-Criteria Evaluation / Weighted Linear Combination (MCE/WLC) method is one of the more practical approaches for addressing multi-factor spatial heterogeneity, and has been widely used to integrate accessibility criteria into a composite suitability surface for planning decision-support.

2.1.1 Process of Weighted Overlay and Moran's I Test

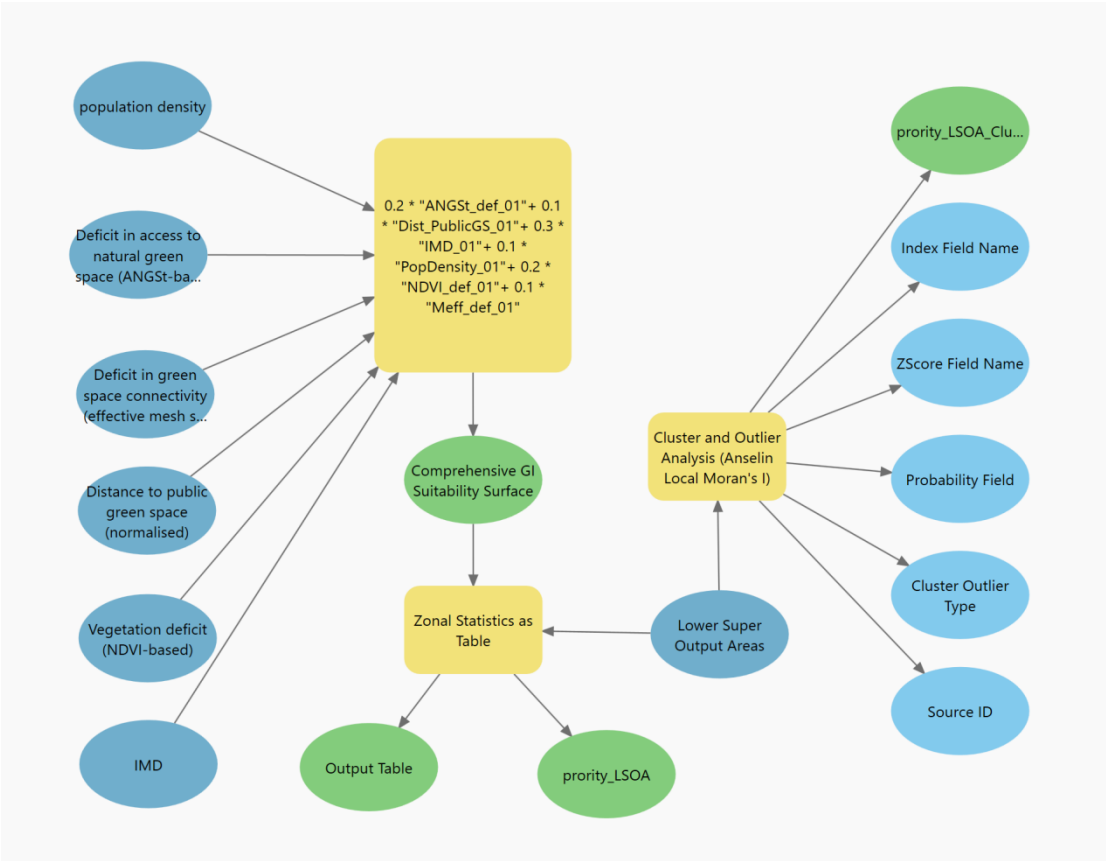

Figure 1. Workflow of weighted overlay and Local Moran's I analysis.

2.1.2 Key Factor Processing: data input and normalization

NDVI:

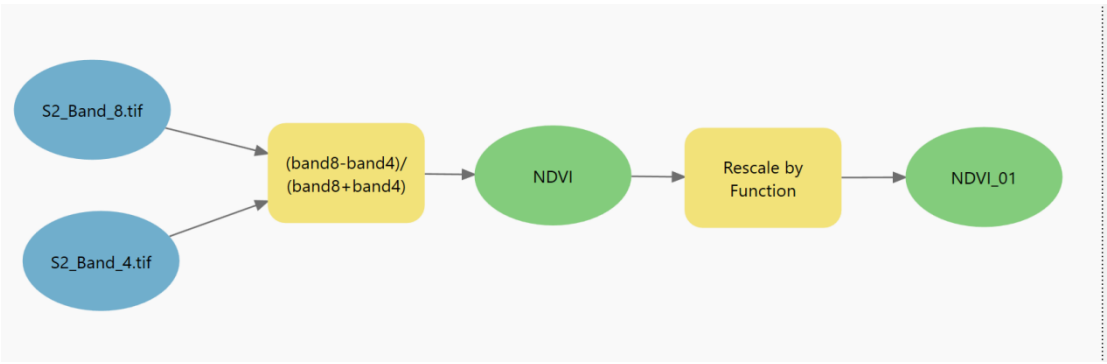

Figure 2. NDVI processing workflow used in the suitability assessment.

ANGSt:

Figure 3. ANGSt accessibility processing workflow.

Distance to public green space:

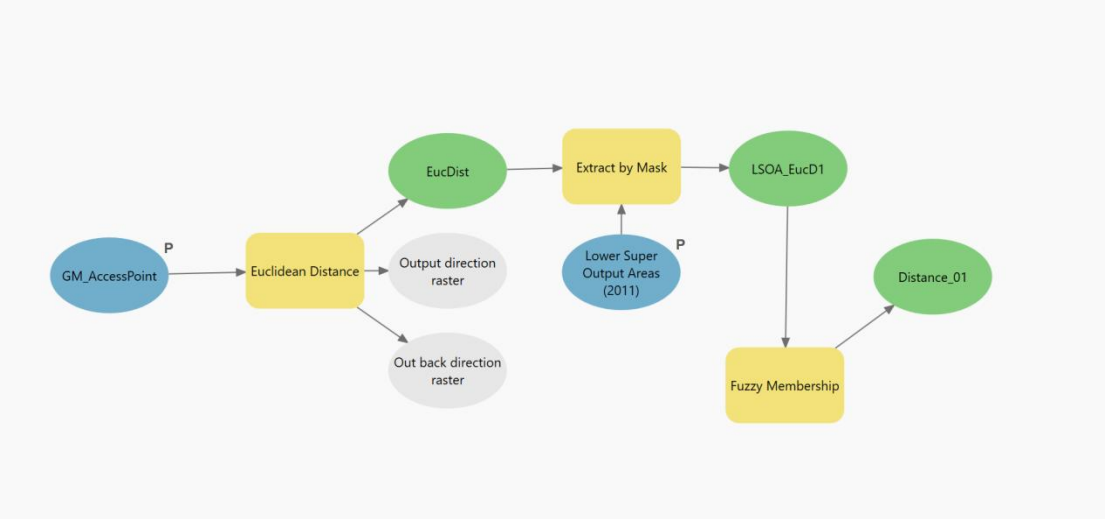

Figure 4. Distance-to-public-greenspace processing workflow.

Meff:

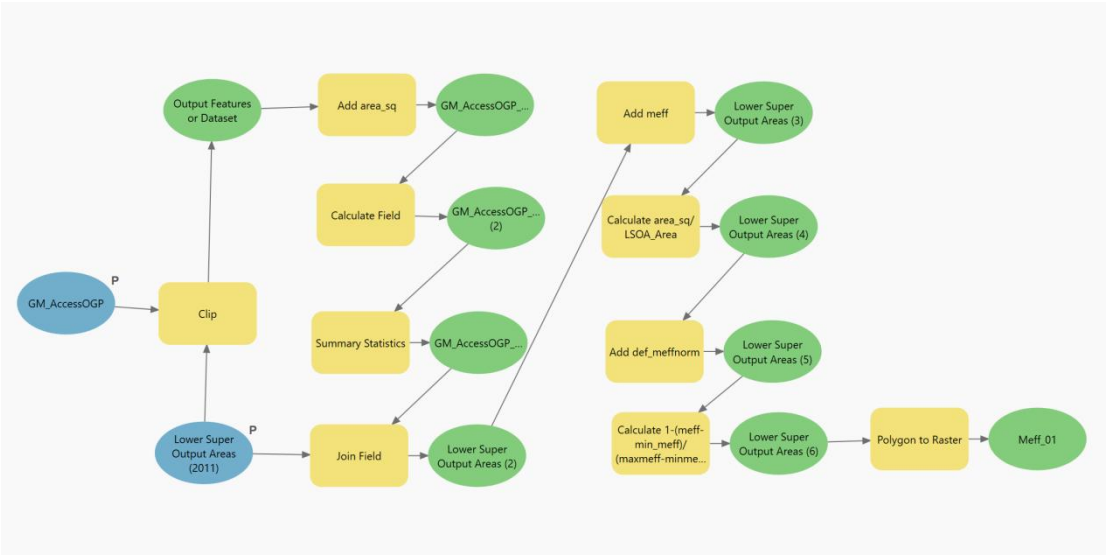

Figure 5. Effective mesh size (Meff) processing workflow.

2.2 Data Sources

The spatial datasets used in this assessment were obtained from a combination of authoritative UK open data sources and satellite-derived products. Administrative boundary and socio-economic data, including Lower Super Output Areas (LSOA) and the Index of Multiple Deprivation (IMD), were sourced from the Office for National Statistics (ONS). Information on public green spaces and designated natural sites was derived from Ordnance Survey Open Greenspace and Natural England datasets. In addition, near-infrared (NIR) and red band satellite imagery were derived from Copernicus Sentinel-2 products provided by datasets were accessed via the course data repository, while land surface temperature (LST) data were exported from the Google Earth Engine cloud platform.

2.3 Criteria Selection and Justification

In this study, a total of six factors, namely population density, Index of Multiple Deprivation (IMD), Normalised Difference Vegetation Index (NDVI), ANGSt-based green space accessibility deficit, distance to public green space, and green space patch connectivity (effective mesh size), were selected for weighting and used to calculate the overall suitability.

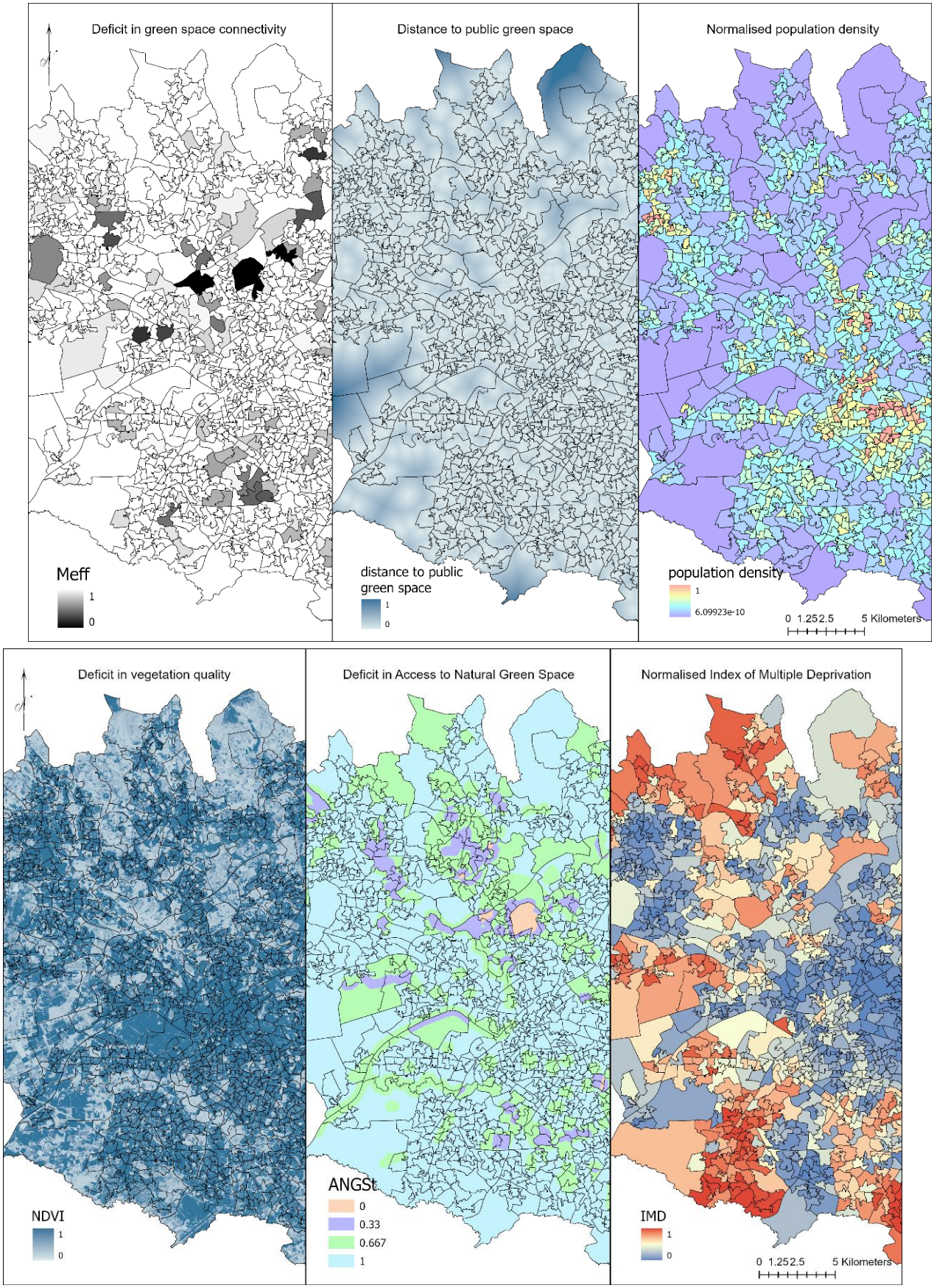

Figure 6. Spatial distribution of key indicators used in the composite suitability analysis.

2.3.1 Social Needs and Equity

Considering the IMD and population density factors addresses the question of who ultimately needs and benefits most from green infrastructure interventions.

The IMD is the UK government's official composite measure of socioeconomic vulnerability, reflecting multiple disadvantages including income, health, education, and living conditions. For less mobile groups such as the elderly or low-income individuals, reliance on green spaces near their residences is even greater. In these areas of heightened social vulnerability, the absence of accessible, high-quality public green spaces exacerbates chronic health risks like obesity, depression, and cardiovascular disease. Furthermore, these populations are more susceptible to climate change threats (e.g., heatwaves), and the lack of green spaces further amplifies their health risks (Dennis et al., 2020). Simultaneously, population density serves as a measure of the potential beneficiary population for green space development, making it a crucial indicator for maximizing the social benefits of green infrastructure interventions.

Therefore, in this study, IMD was selected as the primary weighting factor for the social dimension, assigned a weight of 0.3. High population density values represent the intensity of intervention demand, serving as a secondary factor for the social dimension with a weight of 0.1.

2.3.2 Spatial Accessibility and Coverage

ANGSt and distance to public green factors were considered.

ANGSt is a widely adopted UK government standard for assessing natural green space accessibility planning, explicitly stipulating that residents should have access to natural green spaces of varying sizes under the following conditions:

> *People must have an accessible natural greenspace:*
>
> *- of at least 2 hectares in size, no more than 300 meters (5 minutes walk) from home;*
>
> *-at least one accessible 20-hectare site within two kilometers of home; one accessible 100-hectare site within five kilometers of home;*
>
> *-and one accessible 500-hectare site within ten kilometers of home;*
>
> *plus a minimum of one hectare of statutory Local Nature Reserves per thousand population. (Natural England, 2010)*

Access to public greenspace further emphasizes the universality and multifunctionality of green space services. Therefore, in this study, ANGSt most directly reflects whether service provision meets standards. As the core factor in the accessibility dimension, it is assigned a weight of 0.2. Distance to Public Green, serving as an auxiliary factor, is assigned a weight of 0.1.

2.3.3 Green Space Ecological Quality

Research indicates that in certain communities, fragmented green spaces characterized by "proximity but small size and poor quality" fail to deliver health and ecological benefits comparable to those of large or connected green spaces (Dennis et al., 2020).

So this assessment takes NDVI as the primary indicator with a weighting of 0.2 to measure existing vegetation growth conditions and ecological service capacity, using the following algorithm:

$$NDVI = \frac{rho_{\{NIR\}} - rho_{\{Red\}}}{rho_{\{NIR\}} + rho_{\{Red\}}}$$

Green space connectivity was quantified by the effective mesh size (Meff), which represents the size of an equivalent continuous green space and can be interpreted as the probability that two randomly chosen points fall within the same patch, multiplied by the total landscape area (Dennis et al., 2020). In practice, Meff was calculated as:

$$m_{eff} = \frac{1}{A}\sum_{i=1}^{n} a_i^2$$

where A is the total area of the LSOA and a_i denotes the area of individual green space patches.

Meff measures the degree of green space fragmentation and serves as a supplementary indicator for green space structure and quality, with a weight of 0.1.

2.3.4 Other

The project "Site Selection for Green Roof Suitable Areas" incorporates mitigation of the urban heat island effect, with additional consideration given to the Land Surface Temperature (LST) factor. The LST factor is critical for assessing the mitigation potential of the urban heat island (UHI) effect. This assessment utilized Landsat 9 data processed via remote sensing imagery on the GEE platform to derive 30m resolution land surface temperature data for the Manchester region from June to August.

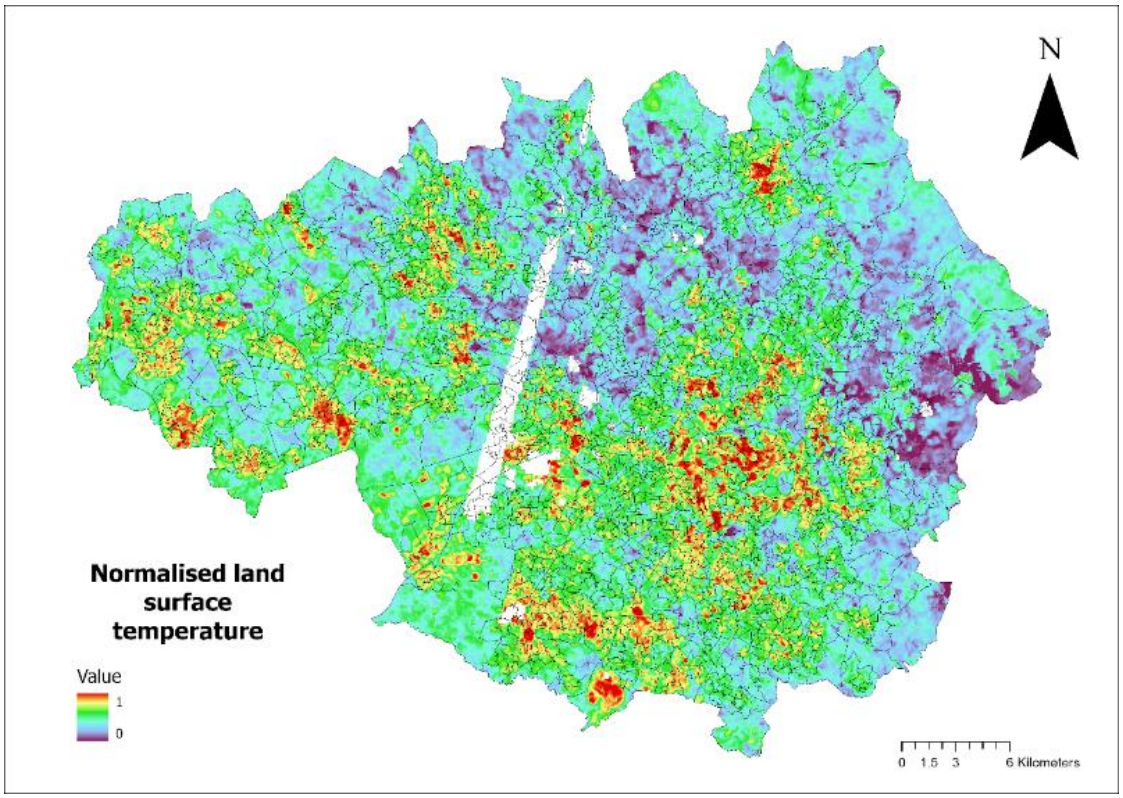

Figure 7. Land surface temperature surface for Greater Manchester derived from Landsat 9 imagery (June-August).

3. Results and Discussion

3.1 Comprehensive Suitability Map

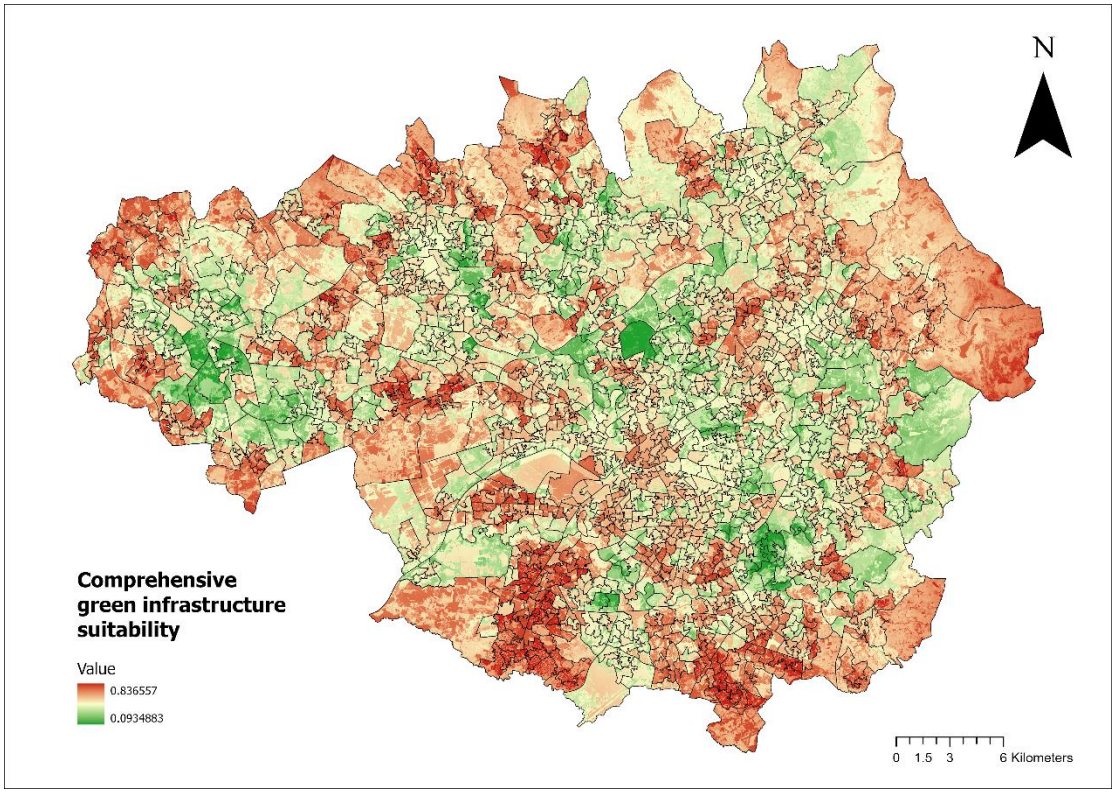

Figure 8. Comprehensive suitability map for green infrastructure interventions in Greater Manchester.

The comprehensive suitability map was generated through a weighted combination of six indicators. Red and green colours are used to illustrate the spatial distribution of comprehensive green infrastructure suitability across the study area, where higher values indicate greater suitability for green infrastructure interventions.

From a visual inspection of the map, higher suitability values are concentrated In the peripheral areas of the Manchester region, particularly in the southern area, the northeastern area and the northwestern area. In contrast, lower suitability values are more commonly observed in the center of Geater Manchester.

This pattern suggests that those areas with large socio-environmental deficits and low access to green space have higher suitability scores, reflecting a stronger potential need for green infrastructure interventions.

3.2 Top 10% of high-suitability areas

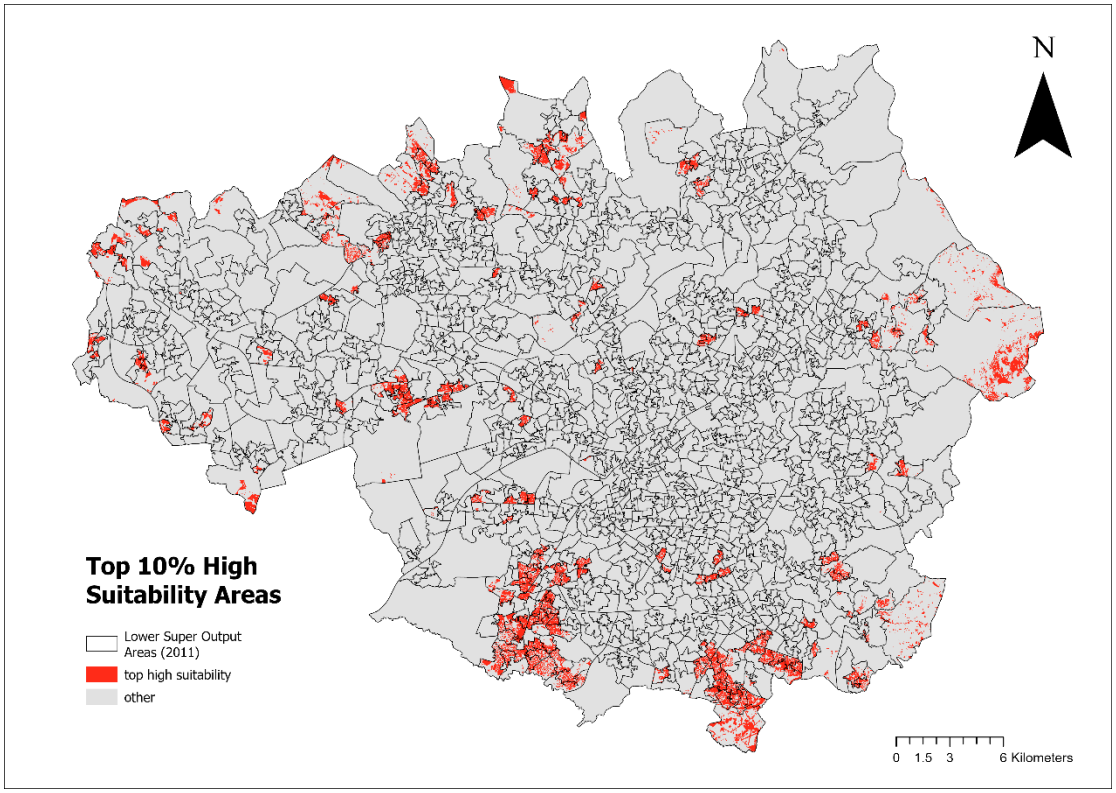

Figure 9. Top 10% high-suitability areas identified from the composite suitability surface.

Based on the composite suitability surface, further calculations highlighted the top 10% of high-suitability areas within the study region. These areas represent locations with the highest composite scores derived from the weighted integration of six indicators, thereby identifying them as priority zones for potential green infrastructure interventions.

High suitability areas exhibit clustered spatial distribution rather than random dispersion, particularly evident in peripheral communities, especially in the southern and eastern parts of the Manchester region. In contrast, relatively fewer top-ranked areas were identified within the central urban core.

3.3 Spatial Statistical Validation

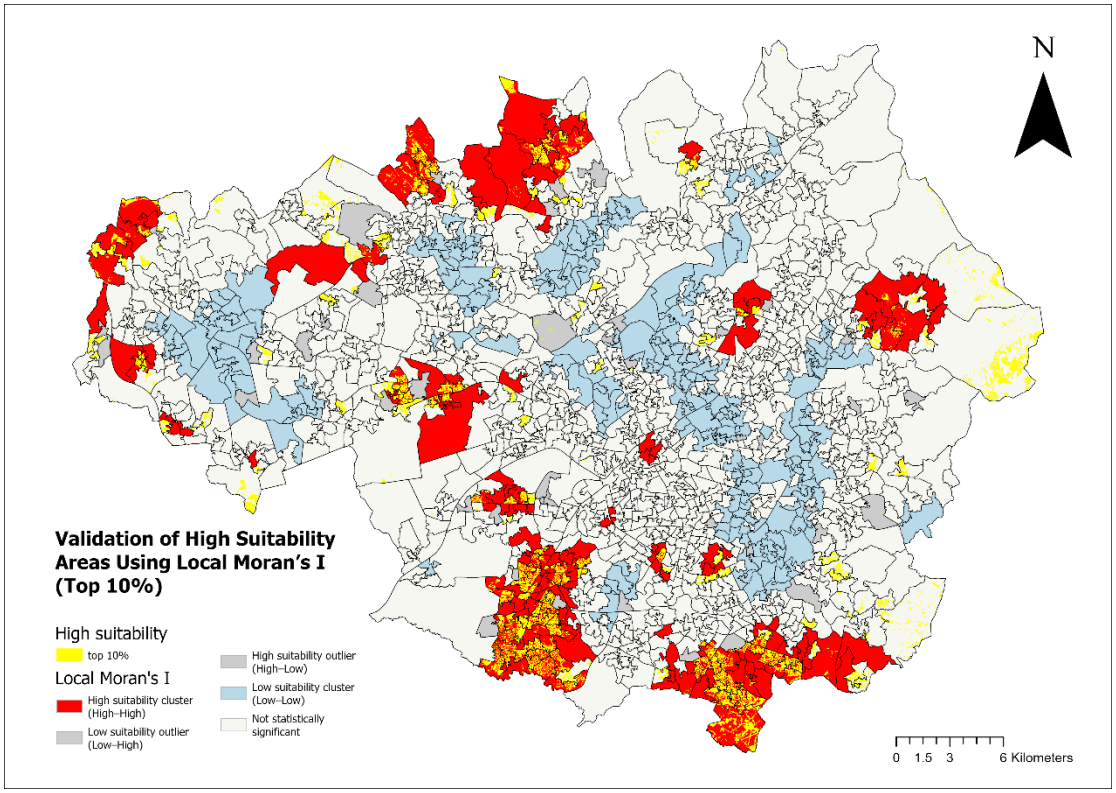

Figure 10. Local Moran's I (LISA) cluster map for the composite suitability surface.

The figure above displays the results of Local Moran's I (LISA) analysis for the composite green infrastructure suitability surface. The red areas indicating high–high clustering reveal that high suitability values are spatially surrounded by similar high values. In contrast, the blue areas representing low–low clustering denote consistently low suitability zones, primarily distributed in the urban core. A limited number of spatial outliers (high–low and low–high) were also observed, reflecting local deviations from the surrounding neighborhood environment.

Overall, the results demonstrate that the comprehensive suitability surface exhibits significant spatial autocorrelation rather than a random spatial pattern. Furthermore, comparing the Top 10% high suitability site selection results with the high–high clustering zones reveals high spatial consistency between the two. This partially validates the statistical robustness and planning effectiveness of the identified priority site selection areas.

3.4 Scenario-based Site Selection

3.4.1 Site selection of rooftop garden

Figure 11. Candidate rooftop garden sites selected within high-suitability areas.

This map overlays comprehensive high-suitability target areas (see above) with building outline data at the LSOA scale. The weight of the land surface temperature factor was recalibrated in the WLC process in a way which better reflects the potential benefits that rooftop greening can bring to mitigating thermal environments. This allowed a series of specific building units to be identified as potentially suitable for the implementation of rooftop green infrastructure interventions.

This map depicts the local example in the study area, where the dark green polygons represent buildings situated in high suitability zones that are characterized by potential conditions for rooftop greening intervention.

3.4.2 Site selection of Largest Community Garden

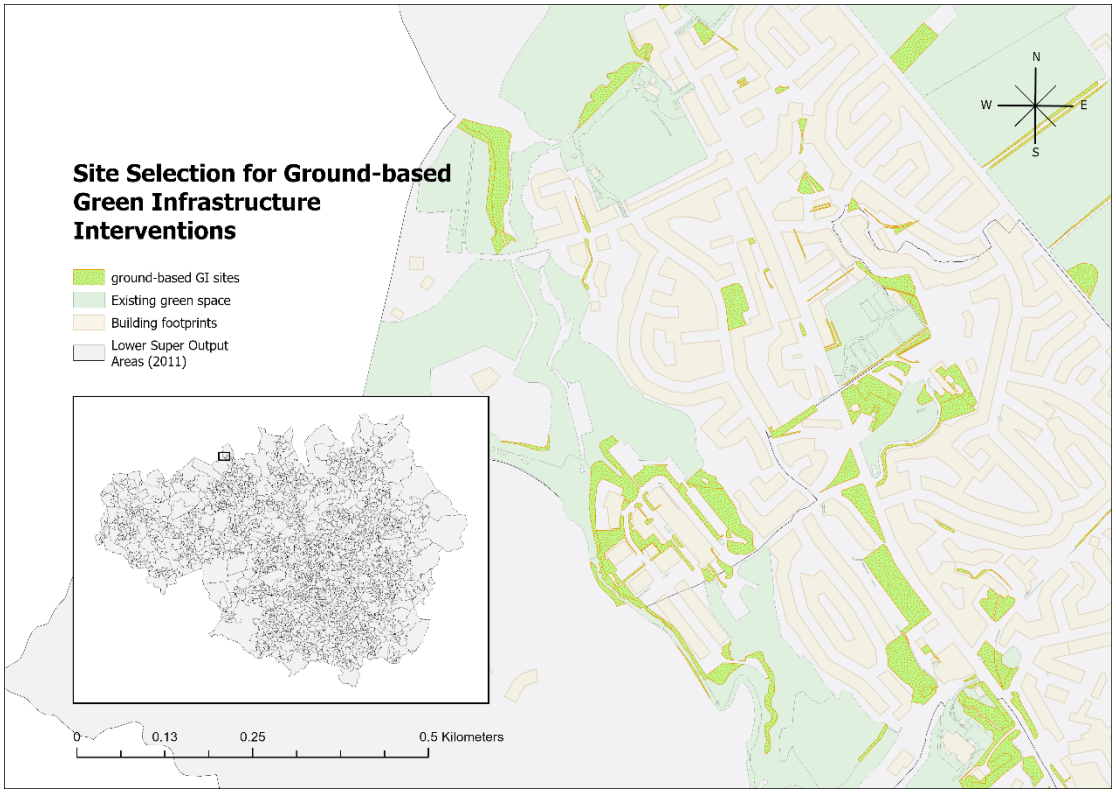

Figure 12. Candidate site for the largest community garden demonstration area.

The site selection of community gardens is based on the results of the comprehensive green infrastructure suitability analysis and is screened in combination with spatial feasibility and social needs. It first extracts the spatial units from the "available area for ground greening", which intersects with the high-suitability sub-areas of the comprehensive evaluation, so as to ensure the basic implementation conditions. Then, it introduces the IMD and ANGSt to preferentially identify the areas with high social needs and insufficient green space supply. Finally, it screens the candidate plots in conjunction with land use type and plot scale, and selects the plots of medium size, which is convenient for community management and relatively easy to access, as the demonstration implementation site of community gardens.

3.5 Critical Evaluation

3.5.1 Methodological limitations

Weight allocation relies to some extent on the researcher's subjective judgment. In this assessment, the ANGSt factor primarily evaluates whether a region meets national-level standards for natural green space accessibility, emphasizing the normative provision of legally designated or quasi-legal natural green spaces (e.g., LNR). Conversely, the public green space distance factor within the IG indicator focuses on depicting the actual accessibility of public green spaces available for residents' daily use. Theoretically distinct, both metrics reflect deficiencies in green space provision and accessibility at different scales, resulting in conceptual overlap. This overlap may amplify the influence of relevant areas during weighted aggregation, thereby impacting the final weight distribution and overall suitability outcomes.

3.5.2 Data uncertainty and scale effects

This research has some uncertainty in the temporal consistency of the data being used. The ANGSt metric uses Local Nature Reserve data from October 2025, and similarly, the Meff uses Open Green Space data from the same period. The IMD dataset, on the other hand, used information published in 2019, and population data relies on ONS 2011 Census paired with Ordnance Survey OpenData. The temporal inconsistency across these datasets means that the model will overlay static socio-economic data with dynamic urban spatial patterns continuously changing. This temporal mismatch might affect the accuracy and real-world application of the results.

Apart from that, during LST calculation, the original Landsat data had a spatial resolution of 30 meters. This study resampled the LST raster into 10 meters to maintain consistency with other high-resolution raster data, such as NDVI. Although this process unifies the spatial scales across different multifactor analyses, the resampling may reduce the authenticity of local temperature variations.

3.5.3 Implications for planning practice

Ground-truthing is absolutely necessary while carrying out some precise planning analyses in ArcGIS. For example, campus land primarily covered by forest must be considered in terms of its actual ecological value and functional use.

GI calculations involving LNR and public green space should consider the dilution effect of large private and institutional green spaces on public green space demand. If the majority of the users in a particular area are already served by enough non-public green space resources, then

the need to add new public green space in that particular area is justified only moderately from an optimal resource allocation perspective.

Consider topographic impedance. Many cities contain large public green parks built on hillsides. Even if a park seems geographically close to residential areas on maps, accessibility assessments relying purely on buffers and Euclidean distances are highly limited. These kinds of parks may primarily provide space for fitness enthusiasts. So, future planning could consider doing "per capita green space exposure analyses" for specific demographics. This would give more precise data support and reference for future planning decisions.

4. Conclusion and Recommendations

This study applies MCE and WLC methods to establish a spatial identification framework for green infrastructure intervention zone prioritization in Greater Manchester, UK. Based on the integration of social, environmental, and spatial indicators, an integrated suitability surface was generated.

The spatial clustering feature of the high-suitability area was verified using Local Moran's I, enhancing the statistical robustness of the site selection results. Based on the abovementioned foundation, this paper further differentiated GI intervention types and refined regional-scale suitability results into building and plot scales, which demonstrated a potential application of integrated suitability analysis in planning practice.

Considering the limitations in such aspects as weighting parameters, data timeliness, and real-world implementation, this framework provides a useful reference for spatial decision-making in the name of equitable and refined urban green infrastructure planning.

5. References

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Dennis, M., Armitage, R.P., James, P., 2016. Appraisal of social-ecological innovation as an adaptive response by stakeholders to local conditions: Mapping stakeholder involvement in horticulture orientated green space management. Urban Forestry & Urban Greening 18, 86–94. https://doi.org/10.1016/j.ufug.2016.05.010

Natural England (2010) Nature Nearby: Accessible Natural Greenspace Guidance. Sheffield: Natural England.

Drobne, S. and Lisec, A. (no date) 'Multi-attribute Decision Analysis in GIS: Weighted Linear Combination and Ordered Weighted Averaging'.

6. Acknowledgements

Contains Ordnance Survey data © Crown copyright and database right (2024).
Other datasets were obtained from UK Government Open Data, Copernicus Sentinel-2, and Google Earth Engine.

7. Appendix – Geospatial Metadata

Layer name	Type	Description	Data sources from	link
Lower Super Output Area(2011)	Vector (Polygon)	Administrative boundary units used for aggregating socio-economic and environmental indicators at neighbourhood level.	Course data repository	
LNR_GM	Vector (Polygon)	representing areas of legally designated high-quality natural green space. This layer was used to assess access to high-value natural environments and to support the ANGSt-based green space accessibility analysis.	Published by Natural England, Last updated: 09 October 2025	Local Nature Reserves (England)
GM_popgrid	Raster	Gridded population density raster for Greater Manchester, representing the spatial distribution of residential population. This layer was used to identify areas of higher population demand and to support population-weighted green infrastructure suitability assessment.	Course data repository	
IMD	Vector (Polygon, attribute-based)	Index of Multiple Deprivation scores at LSOA level, representing relative socio-economic deprivation.	From:Ministry of Housing, Communities & Local Government (2018 to 2021) Published:26 September	https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019
SK_GreenspaceSite	Vector (Polygon)	Public green space site polygons generated for the Greater Manchester study area. The dataset was subdivided into four regional subsets (SK, SD, SJ, and SE) to correspond with the administrative sub-regions of Greater Manchester	OS OpenData,OS Open Greenspace product from ordnance survey	https://www.ordnancesurvey.co.uk/products/os-open-greenspace
SD_GreenspaceSite	Vector (Polygon)			
SJ_GreenspaceSite	Vector (Polygon)			
SE_GreenspaceSite	Vector (Polygon)			
SK_Building	Vector (Polygon)	buliding rootprint polygon,genered for Site selection use. The dataset was subdivided into four regional subsets (SK, SD, SJ, and SE) to correspond with the administrative sub-regions of Greater Manchester	OS Vector Map District product from ordnance survey	https://www.ordnancesurvey.co.uk/products/os-vectormap-district#overview
SD_Building	Vector (Polygon)			
SJ_Building	Vector (Polygon)			
SE_Building	Vector (Polygon)			
SK_AccessPoint	Vector (Point)	Access points to public green spaces used for distance for public green space analysis.The dataset was subdivided into four regional subsets (SK, SD, SJ, and SE) to correspond with the administrative sub-regions of Greater Manchester		
SD_AccessPoint	Vector (Point)			
SJ_AccessPoint	Vector (Point)			
SE_AccessPoint	Vector (Point)			
LST_C_median_2024-06-01_to_2024-08-31	Raster	Median land surface temperature (°C) derived for 2024-06-01 to 2024-08-31 fliter:CLOUD_COVER'< 20, spetial resolution:30m used to represent urban thermal environment.	Google Earth Engine (Landsat-based LST products)	
S2_Band_4	Raster	Sentinel-2 red bands used to calculate NDVI.	Course data repository	
S2_Band_8	Raster	Sentinel-2 near-infrared bands used to calculate NDVI.	Course data repository	